

Project Workflow: Week 1

Phase: Project Planning & Concept Finalization

1. Objective

The primary goal of Week 1 was to establish a robust project foundation. This included:

- Defining the project scope and functional boundaries.
- Selecting an optimal technology stack for scalability and performance.
- Conceptualizing the UI/UX design to align with professional standards.

2. Detailed Task Breakdown

- Feature Identification: Conducted research on standard scientific calculators to finalize the core 15+ essential functions (Arithmetic, Trigonometric, Logarithmic, and Exponential).
- Technical Stack Selection: Selected **Java Swing** for its mature component library, stability, and platform independence.
- UI/UX Design: Created wireframes for a 7x5 button grid and established a dark-themed visual identity (Background: #18181B) to ensure a modern, ergonomic user experience.
- Architectural Setup: Designed a modular, single-class architecture (extending JFrame) to ensure clean code separation between UI construction and event handling logic.

3. Deliverables

- Project Scope Document: A formal outline of the calculator's features and limitations.
- UI Wireframe Sketches: Conceptual layouts for interface components.
- Color Palette Definition: A finalized AWT color schema for the application.
- Project Timeline: A structured 10-week development roadmap.

4. Tools & Resources

Development Environment:

- Java JDK 11+
- IntelliJ IDEA/VS Code.

Reference Materials: Java Swing and AWT